MIL-C-43613C
14 April 1986
SUPERSEDING
MIL-C-43613B
28 December 1979

# MILITARY SPECIFICATION

## CAN, WATER, MILITARY: PLASTIC, 5-GALLON CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This document covers a 5-gallon capacity, rectangular, plastic, military water can, with carrying handle.

## 2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| L-P-390 | - Plastic, Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium, and High Density) |
| PPP-F-320 | - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-9-248 - Can, Water, Military, Plastic, 5-Gallon Capacity; Assembly
2-9-249 - Can, Water, Military, Plastic, 5-Gallon Capacity; Cap Assembly
2-9-285 - Adapter, Assembly and Details
10-1-487 - Can, Water, Plastic, 5-Gallon Capacity; Packing for Level B Shipment

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations, unless a specific exemption has been obtained.



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## 3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

* 3.2 Federal Food and Drug Act certification. Prior to the approval of the first article, if one is submitted, or prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative, that the plastic for the water can and the antioxidants, light stabilizers, or other substances incorporated in the plastic he proposes to supply under this specification meet the requirements of 21 CFR - Food and Drugs: Indirect Food Additives: Polymers, Subpart B, Part 177, Chapter I, Title 21, Code for Federal Requirements for Contact with Food. Acceptable evidence of meeting the requirements of this document shall be a certified test report from a recognized independent testing laboratory, acceptable to the Department of the Army Surgeon General. This certification will be required for all acquisitions without exception.

3.3 Materials.

* 3.3.1 Polyethylene, high density. The plastic body shall consist of a high density polyethylene copolymer resin conforming to type III, class H of L-P-390 plus a light stabilizer, and requirements of 3.2 except that the basic resin prior to adding the light stabilizer shall have a density of not less than 0.954 and not more than 0.956, and a melt index of not more than 0.3.

* 3.3.2 Polyethylene, low density. The complete plastic cap and strap shall be low density virgin polyethylene conforming to type III, class L of L-P-390 plus a light stabilizer and requirements of 3.2.

* 3.4 Design and construction. The design and construction of the water container shall be as shown on the drawings specified in 2.1 and as specified herein. Silicone-type mold release agents shall not be used with polyethylene. Mold release agents which are not active stress-cracking agents for polyethylene shall be used. A certificate of compliance shall be furnished stating that the mold release agents comply with this requirement (see 4.5). A certificate of compliance shall be furnished stating that the material with the light stabilizers used on the complete cap and body meet the test requirements of 4.6.4. The male and female threads shall be fully formed and free from flash or thread misalignment at the parting line. The body shall be fabricated from the high density polyethylene specified in 3.3.1, and shall consist of a bottom, four sides, and top with handle and a body neck and shall be formed by blow molding to give a one piece construction. The body walls shall be a minimum of 0.100 inch thick, except that the thickness of the four top outside curved surfaces of the handle may be 0.075 inch minimum;



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however, the 0.075-inch minimum thickness shall not extend over an area greater than 0.25 inch in diameter. The body neck shall have a thickness of not less than 0.151 inch in all areas below the area finished to form the liquid seal. The body walls shall not crack or break when tested as specified in 4.6.2. The upper surface of the body neck that comes in contact with the tapered sealing surface on the inside of the cap to form a seal shall be flat and smooth and may be out of round, in a plane parallel to the top surface of the body neck, provided that the fit or water tightness is not impaired. The interior surface of the body shall be smooth. The interior rounded surface area where the body and the neck join shall have a uniform texture having no rough areas, surface abnormalities, surface deformations, or surface irregularities exceeding those on the standard sample in magnitude and frequency (see 6.4), when compared with the standard sample. The finished water can shall not leak when tested as specified in 4.6.1. The cap assembly components, except the retaining pin, shall be injection molded and fabricated of the material specified in 3.3.2. The retaining pin shall be fabricated from the material specified in 3.3.1 and may be heat sealed to the breather stud cap. The water can cap shall lock in the closed or tightened position by means of a tab on the water can cap assembly, which is engaged to the raised tab located at the base of the body pouring neck on the body. The breather stud cap shall be secured to the breather stud in the closed or tightened position by means of a retaining pin inserted through both the breather stud and breather stud cap. The breather stud shall be free and clear of all mold flashes. The caps or cap areas shall not crack or break when the water can is tested in accordance with 4.6.2. The plastic caps shall not crack when tested in accordance with 4.6.3. The upper surface of the cap pouring spout neck, that comes in contact with the tapered sealing surface on the inside of the pouring spout cap to form a seal, shall be flat and smooth and may have an out of roundness in a plane parallel to the top surface of the pouring spout, provided that the fit or water tightness is not impaired.

* 3.5 Color. The through color of the material for the body, caps, and breather after molding shall approximate color No. 30279 (sand, lusterless) of FED-STD-595. The complete cap and breather shall match the shade of the body.

3.6 Finish. The exterior surfaces of the water can and the cap assembly shall have a smooth and lusterless finish throughout. All surfaces shall be free of dirt, dust, and foreign matter inclusion. The finish shall be produced from a die that has cavity surfaces finished in vapor blast, water hone, or similar satin finish, but not by the application of any protective coating, lacquers, or other materials. The inside of the cap assembly, including threaded portion, shall possess a uniformly high luster throughout as imparted by a polished die.



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3.7 Marking. The water can shall be permanently marked "WATER" by an embossed molded-in process. The water can shall be marked by an embossed molded-in process, with U.S., year of manufacture, and contractor's name or trademark. The container cap shall be marked with raised characters by an embossed molded-in process with the manufacturer's name, or trademark.

3.8 Workmanship. The water can shall be clean, and free from flash, bubbles, cuts, tears, holes, burns, breaks, cracks, pinholes, pits, warpage, blisters, scratches, or any weld marks or sink marks, except that mold parting lines are acceptable. The water can shall not show any signs of discoloration, surface deterioration, roughness, orange peel, or grease.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary, to assure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).



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4.3 First article inspection. When a first article is required, it shall be examined for the defects in 4.4.2.1 and the specified dimensions, and tested in accordance with 4.6.1, 4.6.2, 4.6.3, and 4.6.4. Failure to pass any test shall be cause for rejection of the first article. The presence of any defect listed in 4.4.2.1 or any dimension not within specified requirements shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 In-process inspection.

4.4.2.1 In-process examination. Examination shall be made of the fabrication processes in table I to establish conformance with specified requirements. Whenever noncompliance is noted, correction shall be made to the applicable process and all items processed.

TABLE I. In-process examination

| Requirement | Reference Paragraph |
|---|---|
| Blow molding of can body | 3.4 |
| Injection molding | 3.4 |
| Cavity surfaces of finished die | 3.6 |

4.4.3 End item inspection. The inspection lot shall be all water cans offered for inspection at one time. The sample unit shall be one complete water can.

* 4.4.3.1 Visual examination. Examination shall be made of the end item for defects listed in table II. The inspection level shall be II, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total defects.



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TABLE II. Classification of visual defects

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| Finish | Color not as specified | 101 | |
| | Exterior surface of body, cap, or strap not smooth and not lusterless throughout (except any die parting lines) | 102 | |
| | Inside surface of cap, including threaded portion, not smooth | | 201 |
| | Inside surface of cap, including threaded portion, not of uniform high luster | | 202 |
| Construction and workmanship (applicable to all assemblies and components) | Component missing | 103 | |
| | Any bubbles, cut, tears, holes, flash, burns, breaks, cracks, pinholes, pits, warpage, blisters, scratches, welds or sink marks, or not clean or smooth | 104 | |
| | Any discoloration, surface deterioration or roughness, foreign inclusion, or orange peel | 105 | |
| Body | Sealing surface of lip is not flat, smooth, and free from sink marks or depressions | 106 | |
| | Interior of neck opening not smooth | 107 | |
| | Interior surfaces of the body not smooth | 108 | |
| Cap and strap (large cap or spout cap as applicable) | Sealing surface in cap is irregular, not smooth, or contains high or low points | 109 | |
| | Strap locking tab does not engage with tab on body in closed position | 110 | |
| | Cap does not screw on body neck or pouring spout | 111 | |



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TABLE II. Classification of visual defects - Continued

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| Cap and strap (large cap or spout cap as applicable) (continued) | Cap binds when screwed on body neck or pouring spout | | 203 |
| | Strap does not engage in retention groove | | 204 |
| | Breather air vent holes not clear | 112 | |
| Threads | Threads in caps, body neck, pouring spout, or breather stripped | 113 | |
| | Threads not fully formed, not free from flash, or misaligned at parting line | 114 | |
| Marking | Missing, incomplete, not legible, not specified type or size, or not in proper location | | 205 |

4.4.3.2 Dimensional examination. Examination shall be made of the end item to determine compliance with those dimensions annotated with an asterisk on Drawing 2-9-248. The inspection level shall be S-2. Any nonconformance with other specified dimensions shall be classified as a defect, and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.3.3 Testing of the end item. The end item shall be tested as specified in 4.6.1, 4.6.2, and 4.6.3. The inspection level shall be S-3, and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4.4 Packaging inspection. An inspection shall be made to determine that packing and marking comply with section 5 requirements. Defects shall be as specified in table III. The lot size shall be the number of palletized loads offered for inspection at one time. The sample unit shall be one palletized load fully packaged. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 4.0.



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TABLE III. Packaging inspection

| Examine | Defect |
|---|---|
| Materials | Any component missing or damaged |
| Workmanship | Inadequate application of components such as missing strapping or loose strapping |
| Contents | Number of water cans per bundle is more or less than required |

4.4.5 Palletization inspection. An inspection shall be made to determine that palletization complies with section 5 requirements. Defects shall be as indicated in table IV. The lot size shall be the number of palletized unit loads submitted for inspection at one time. The sample unit shall be one palletized load. The inspection level shall be S-1, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

TABLE IV. Palletization inspection

| Examine | Defect |
|---|---|
| Finished dimension | Length, width, or height exceeds specified maximum requirement |
| Palletization | Pallet pattern not as specified<br>Load not bonded with required straps specified |
| Weight | Exceeds maximum load limits |
| Marking | Omitted; incorrect; illegible; of improper size or location |



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* 4.5 Certification examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the document requirements shall be examined and validated for mold release agents and light stabilizers (see 3.2 and 3.4).

4.6 Test methods.

4.6.1 Leakage test. The water can shall be filled with water. The cap shall be screwed down and tightened, and the water can shall be allowed to remain on its bottom for 10 minutes and examined for compliance with 3.4. The water can shall then be placed cap down for not less than one-half hour and examined for leakage. To determine compliance with 3.4, any evidence of leaking shall constitute failure of this test.

4.6.2 Drop test. The water can shall be filled with 5 gallons of water. The filled water can shall be conditioned for 24 hours at 0°F. The conditioned water can shall then be dropped from a height of 6 feet onto a solid concrete horizontal surface in such a manner that the base of the container strikes squarely against the surface to determine compliance with soundness requirements in 3.4. After the above drop test has been accomplished, the conditioned water can shall be allowed to regain room temperature and then examined for leakage. Any evidence of leaking shall constitute failure of this test.

4.6.3 Environmental stress cracking tests for caps. The neck threads on the body and the cap threads, including area between the base of threads and sealing ring, shall be thoroughly coated with full strength Igepal CO-630 (see 6.5). The cap shall be screwed down on the body and tightened to a torque of not less than 15 foot pounds by the use of a torque wrench utilizing the adapter shown on Drawing 2-9-285. The container shall then be placed in an upright position in a circulating air area at a temperature of 140° ± 2°F. After 30 hours exposure, the container shall be removed from the oven and the cap examined for cracks. If the cap shows no cracks, recheck the torque. If it is less than 15 foot pounds, retighten the cap to a minimum of 15 foot pounds and reexamine the cap immediately for cracks. The pouring cap shall be screwed to the pouring spout and the breather shall be open during the period of the test. Any visible evidence of cracks or cracking shall constitute failure of this test.

* 4.6.4 Light stabilizer tests for materials used in the body and complete cap. Plastic samples (compression molded plaques) shall be exposed to a 6,000 watt Xenon Arc Weatherometer for 130, 24-hour days, straight running time. There shall be four one-half hour moisture cycles a day. At the completion of the tests, the samples shall show no signs of deterioration or degradation.



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* 5. PACKAGING

5.1 Packing. Packing shall be level A or B, as specified (see 6.2).

5.1.1. Level A packing. Seven water cans shall be packed as shown on Drawing 10-1-487.

5.1.2 Level B packing. Seven water cans shall be packed as shown on Drawing 10-1-487.

5.2 Palletization. Water cans, packed as specified, shall be prepared in a unitized pallet load in accordance with load type VIII of MIL-STD-147, except that the overall height of the load shall not exceed 43 inches. The palletized load shall be provided with a fiberboard cap, with horizontal separator not required, and bonded with primary and secondary straps and horizontal straps centered in each course. The fiberboard cap shall be fabricated with V2s of PPP-F-320.

5.3 Marking. In addition to any special marking required by the contract or purchase order, palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The water cans specified herein are used for supplying potable water to personnel of the Armed Services in zones I through VI but not in zone VII (Arctic).

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.2, and 6.3).
c. Selection of applicable levels of packing required (see 5.1).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample consisting of one can. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.



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6.4 Standard sample. To determine the degree of smoothness required on the interior surface where the neck and body join (see 3.4), a standard sample showing an acceptable smoothness may be obtained from Director, Technical Operations, Defense Supply Agency, Defense General Supply Center, Richmond, VA 23297.

* 6.5 Commercial source. Igepal CO-630 required in 4.6.3 may be obtained from GAF Corp (FSCM 24299), 1361 Alps Road, Wayne, NJ 07470.

6.6 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue have been made. This was done as a convenience only, and the Government assumes no liability whatsoever for inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-43613C | Can, Water, Military: Plastic, 5-Gallon Capacity |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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